

QUESTION 1 – E-COMMERCE SALES ANALYSIS

An e-commerce company has Customer, Product, Order, and Payment datasets.

Perform data exploration, join the datasets, analyze sales by product/category/location, identify correlations and outliers, perform time-based analysis, and create suitable visualizations. Provide 5 key insights and recommendations.

SOLUTION:

1. Data Exploration

- Load the Customer, Product, Order, and Payment datasets.
- Check:
 - Number of rows and columns
 - Missing values
 - Duplicate records
 - Data types
 - Unique customers/products
 - Minimum and maximum order dates
- Handle missing values and remove duplicates where required.

2. Join the Datasets

Join the tables using common keys:

- Customer → Order: Customer_ID
- Product → Order: Product_ID
- Order → Payment: Order_ID

The final dataset should contain information such as:

Customer_ID, Product_ID, Category, Location, Order_Date, Quantity, Price, Sales, Payment_Method

3. Sales Analysis

Analyze sales based on:

Product

- Find top-selling products.
- Calculate total quantity sold and revenue.

Category

- Compare revenue across product categories.
- Identify the most and least profitable categories.

Location

- Compare sales across cities/states/regions.
- Identify high-performing locations.

Sales by product category

Example visualization for comparing revenue across product categories.

₹0 ₹250K ₹500K ₹750K ₹1M Electronics Clothing Home & Kitchen Books

4. Correlation Analysis

Calculate correlations between numerical variables such as:

- Quantity
- Product price
- Discount
- Sales
- Customer spending

For example, a strong positive correlation between quantity and sales indicates that higher quantities generally generate higher revenue.

5. Outlier Detection

Use Boxplots or IQR method to identify:

- Extremely high-value orders
- Unusually large quantities
- Abnormally high product prices
- Unusual customer spending

Investigate whether outliers are genuine purchases or data-entry errors.

6. Time-Based Analysis

Analyze sales by:

- Day
- Month
- Quarter
- Year

Identify:

- Peak sales months
- Seasonal trends
- Low-sales periods
- Growth or decline in revenue

Monthly sales trend

Example visualization showing how sales change over time.

₹405K ₹450K ₹495K ₹540K ₹585K Jan Feb Mar Apr May Jun

7. Five Key Insights and Recommendations

Key Insight	Recommendation
1. A few products generate most of the revenue.	Maintain sufficient stock and promote top-selling products.
2. Some categories perform much better than others.	Increase marketing for high-performing categories and review weak categories.
3. Certain locations generate higher sales.	Target these locations with personalized offers and advertisements.
4. Sales show seasonal patterns.	Increase inventory and promotions before peak sales periods.
5. Some orders are unusually large or expensive.	Investigate outliers to identify bulk customers, fraud, or data errors.

Final Conclusion

The analysis helps the e-commerce company understand customer behavior, identify profitable products and locations, detect unusual transactions, and recognize sales trends. These insights can be used to improve inventory management, marketing strategies, pricing, and overall revenue.

QUESTION 2 – STUDENT ACADEMIC PERFORMANCE ANALYSIS

A college has Student, Attendance, Internal Marks, Assignment, and Examination datasets. Integrate the datasets and analyze student performance based on department, semester, attendance, and marks. Identify correlations and outliers and visualize performance trends. Provide 5 key findings and recommendations.

1. Data Exploration

Load the **Student, Attendance, Internal Marks, Assignment, and Examination** datasets.

Check:

- Number of students and columns
- Missing values
- Duplicate records
- Data types
- Minimum and maximum marks
- Attendance percentage
- Department and semester distribution

2. Integrate the Datasets

Join the datasets using common keys:

- **Student** → **Attendance**: Student_ID
- **Student** → **Internal Marks**: Student_ID
- **Student** → **Assignment**: Student_ID
- **Student** → **Examination**: Student_ID

The final dataset can contain:

Student_ID, Department, Semester, Attendance %, Internal Marks, Assignment Marks, Exam Marks, Total Marks, Grade

3. Performance Analysis

Analyze student performance based on:

Department

- Calculate average marks for each department.
- Identify departments with highest and lowest performance.

Semester

- Compare average marks across semesters.
- Identify semesters with performance improvement or decline.

Attendance

- Compare attendance percentage with academic marks.
- Identify students with low attendance and poor performance.

Marks

- Calculate total/average marks.
- Identify high-performing and low-performing students.

Average marks by department

Example visualization comparing academic performance across departments.

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4. Correlation Analysis

Calculate correlations between:

- Attendance and examination marks
- Internal marks and examination marks
- Assignment marks and examination marks
- Attendance and total marks

For example, a **positive correlation between attendance and exam marks** indicates that students with better attendance tend to achieve higher marks.

5. Outlier Detection

Use **Boxplots/IQR method** to identify:

- Extremely low or high marks
- Unusually high attendance
- Very low attendance
- Abnormally high assignment/internal marks

Investigate whether these values represent genuine student performance or data errors.

6. Performance Trend Analysis

Analyze performance based on:

- Semester
- Department
- Attendance groups
- Examination results

Academic performance across semesters

Example visualization showing the change in average marks across semesters.

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7. Five Key Findings and Recommendations

Key Finding	Recommendation
1. Students with higher attendance generally perform better.	Monitor attendance regularly and provide support to students with low attendance.
2. Some departments have higher average academic performance.	Study successful teaching practices and apply them across departments.
3. Internal and assignment marks can influence overall performance.	Provide regular assessments and feedback throughout the semester.
4. Some students have unusually low marks.	Identify academically weak students early and provide mentoring/remedial classes.
5. Student performance changes across semesters.	Track semester-wise trends and introduce additional academic support during weaker semesters.

Conclusion

The integrated analysis helps the college identify academic trends, understand the relationship between attendance and marks, detect weak or exceptional students, and compare performance across departments and semesters. The findings can support early intervention, personalized mentoring, improved teaching strategies, and better academic planning.